

Diwali Sales Analysis – Project Report

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Tools Used: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn

Objective: Analyze customer purchasing behavior during Diwali to build effective marketing and sales strategies.

Executive Summary

This project analyzes Diwali festival sales data to identify key customer segments, understand purchasing patterns, and evaluate product performance. Using Python and exploratory data analysis (EDA), the project uncovers detailed insights across demographics, state-wise performance, product categories, and occupational trends.

The findings help businesses optimize festive marketing strategies, manage inventory more effectively, and target high-value customer segments to maximize revenue during Diwali.

Business Problem / Problem Statement

Despite high transaction volumes during the Diwali festival period, the company lacks clarity on **who** its key customers are and **which customer segments** contribute the most to sales and orders.

The business needs to answer questions such as:

- Which **gender** contributes more to the sales volume and revenue?
- Which **age groups** are the most active buyers during Diwali?
- Does **marital status** influence shopping behavior?
- Which **occupations** have the highest purchasing tendency?
- Which **product categories and states** generate the most orders and revenue?
- What are the **top-selling products** that should be promoted more?

Objective:

To analyze customer demographics, buying behavior, and product performance using Diwali sales data and provide actionable insights to improve marketing strategies, product targeting, and revenue growth.

Dataset Description

The dataset contains Diwali shopping transactions with the following key attributes:

- **User_ID** – Unique customer identifier
- **Gender** – Male/Female
- **Age Group** – Customer age category
- **Marital Status** – 0 (Single), 1 (Married)
- **Occupation** – Customer profession

- **State & Zone** – Customer geographic location
- **Product Category** – Category of purchased item
- **Orders** – Number of units purchased
- **Amount** – Total purchase amount

Data Cleaning & Preprocessing

Steps performed:

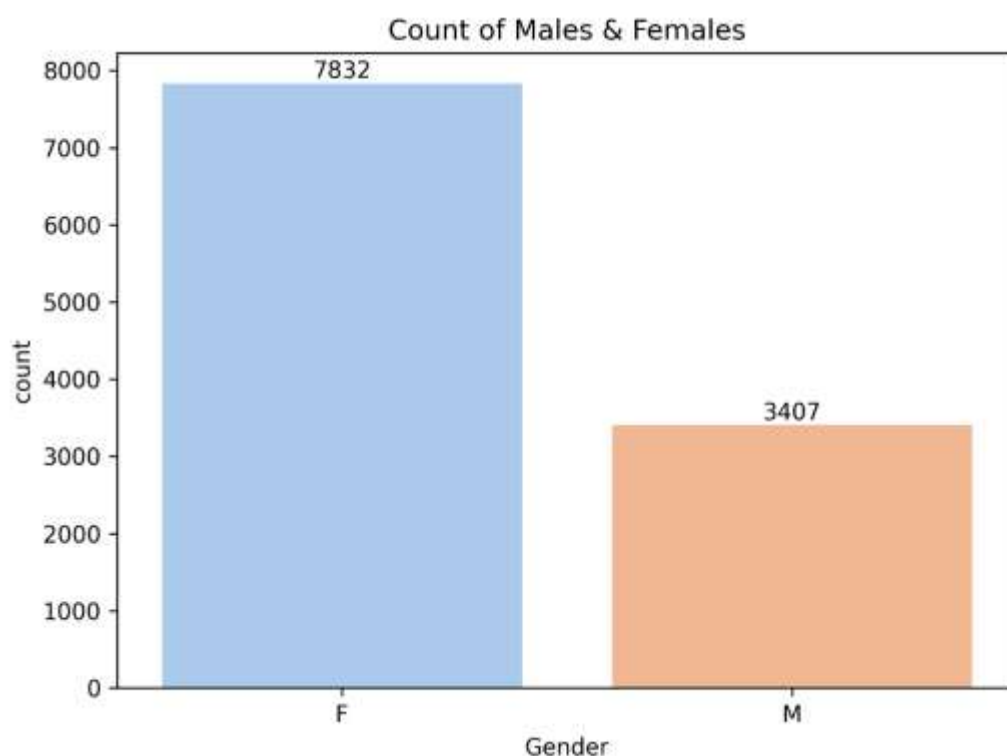
- Removed missing and null values
- Handled incorrect entries and inconsistencies
- Cleaned text formatting (State, Occupation, etc.)
- Converted columns to appropriate data types
- Removed unnecessary columns (Status, unnamed1)
- Created filtered & clean dataset for analysis

Exploratory Data Analysis (EDA)

Below are the charts to include along with insights:

6.1 Gender Analysis

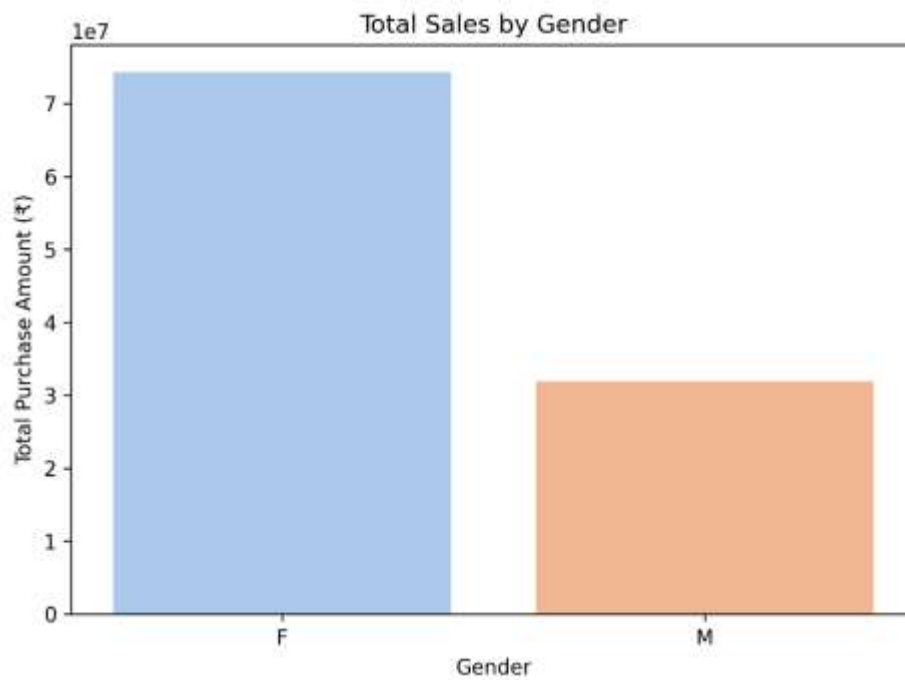
Chart 1: Count of Males & Females



Insight:

Women are the dominant customer segment, accounting for a significantly higher number of orders compared to men.

Chart 2: Total Sales by Gender

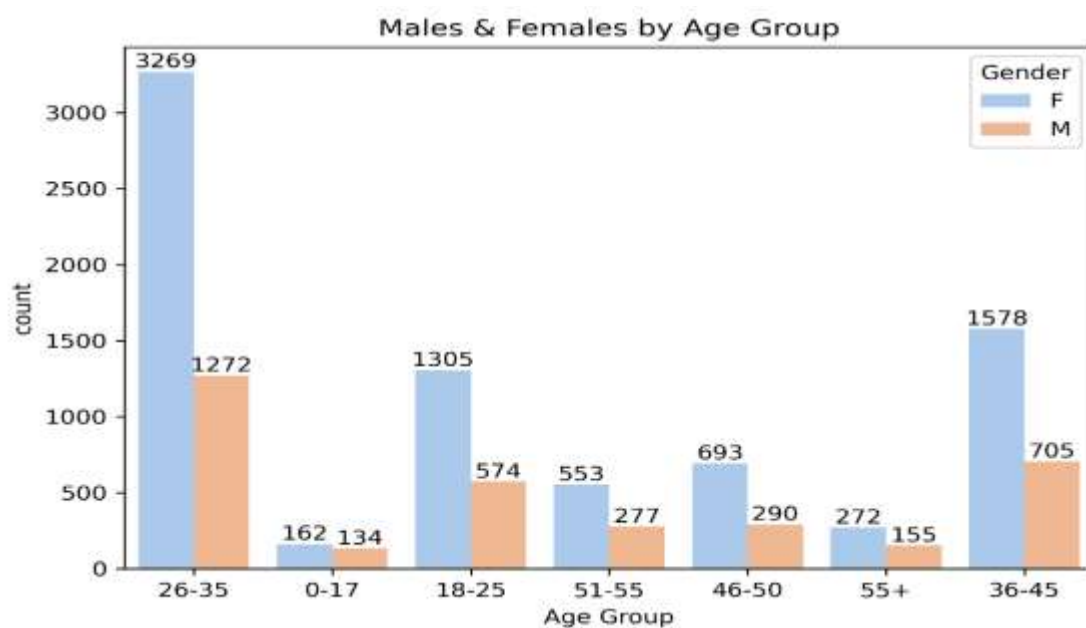


Insight:

Female customers contribute **approximately 75%** of the total sales revenue — making them the most valuable customer segment.

6.2 Age Group Analysis

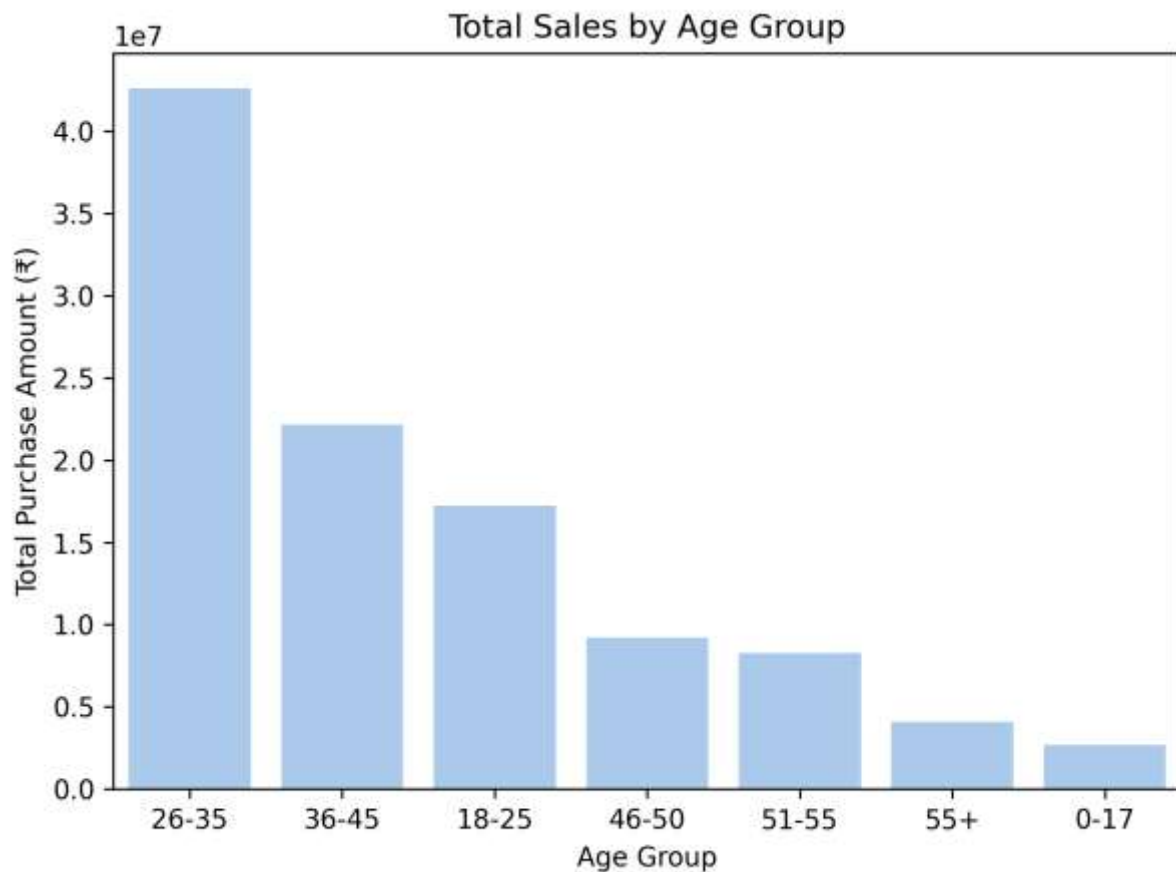
Chart 3: Males & Females by Age Group



Insight:

Females aged **26–35** dominate the Diwali shopping segment with the highest number of purchases.

Chart 4: Total Sales by Age Group

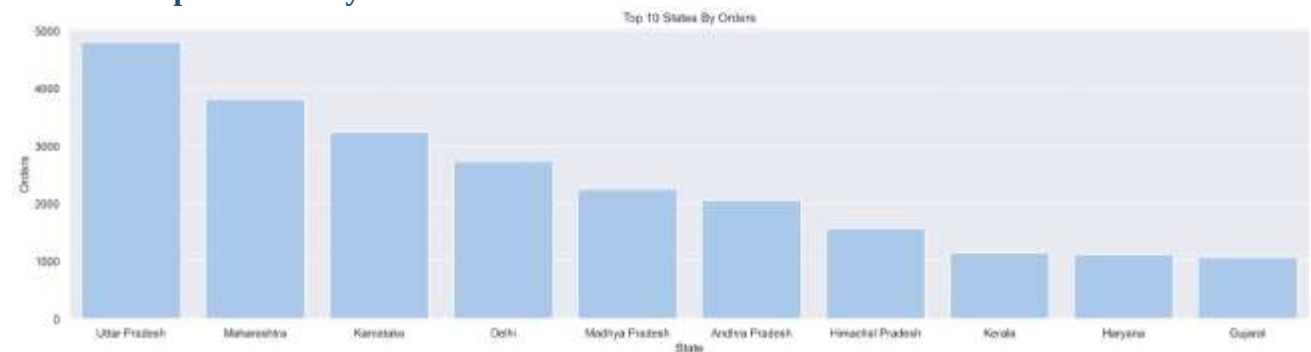


Insight:

The **26–35** age group contributes the highest sales revenue, making them the primary target group for Diwali marketing.

6.3 State-wise Performance

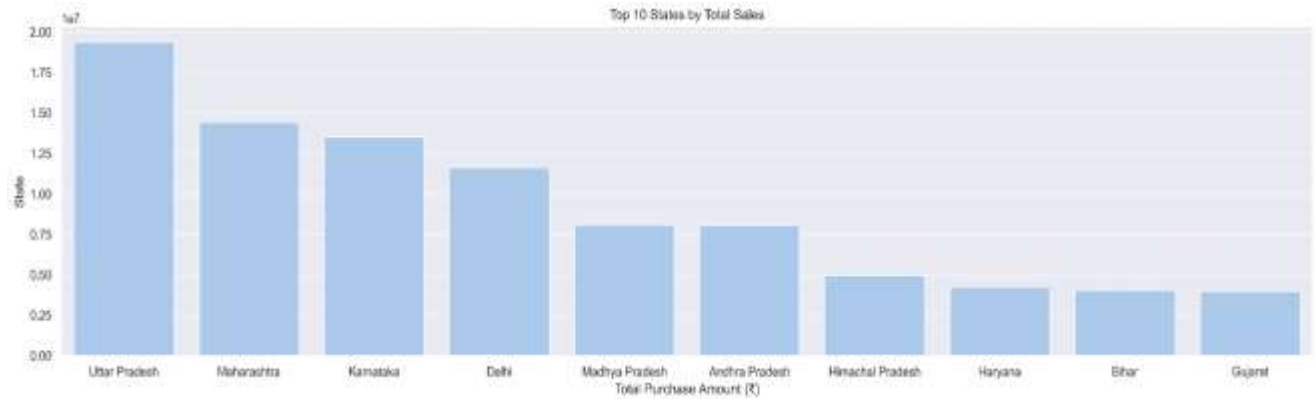
Chart 5: Top 10 States by Orders



Insight:

Uttar Pradesh, Maharashtra, and Karnataka lead in the number of orders, showing strong customer activity

Chart 6: Top 10 States by Total Sales



Insight:

These same states also generate the highest revenue, indicating both high order volume and high purchasing power.

6.4 Marital Status Analysis

Chart 7: Marital Status Distribution by Gender

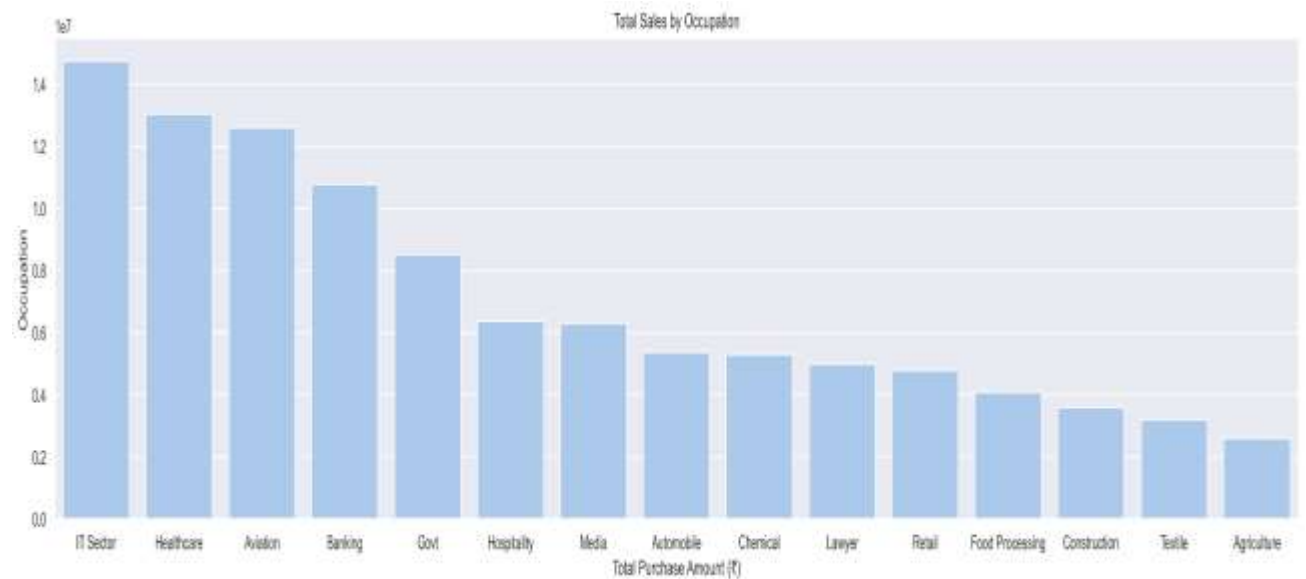


Insight:

Married women generate the highest total purchase value, emerging as the most profitable customer group.

6.5 Occupation Analysis

Chart 8: Total Sales by Occupation

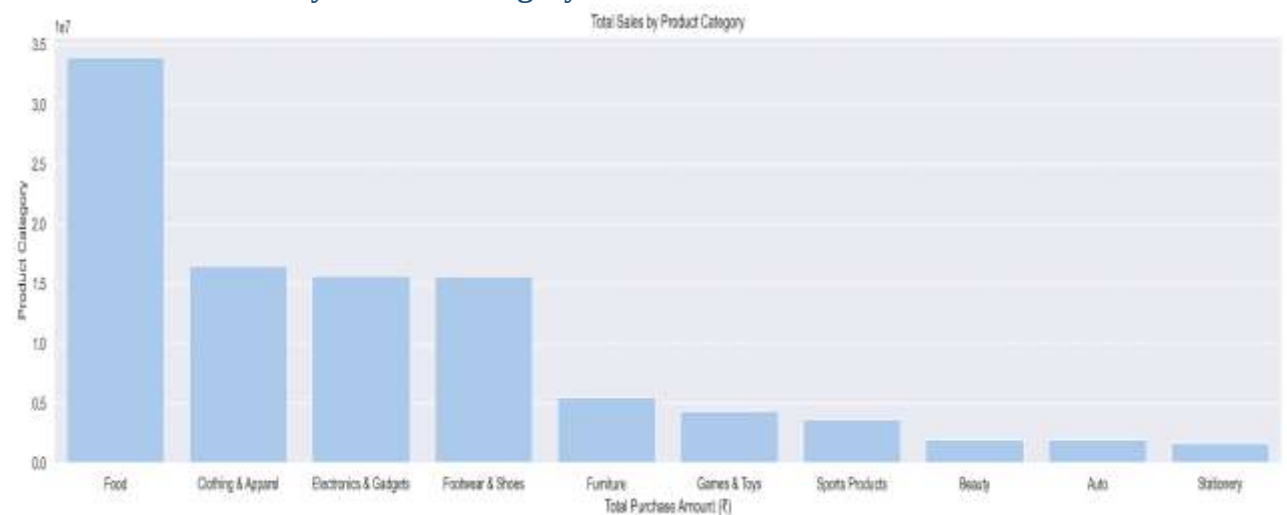


Insight:

Customers working in **IT, Healthcare, and Aviation** show the highest spending, indicating strong purchasing power in these sectors.

6.6 Category Analysis

Chart 9: Total Sales by Product Category



Insight:

Clothing & Apparel, Food, and Electronics are the top-selling product categories, contributing the majority of sales revenue.

Key Insights Summary

1. **Women contribute the majority of total sales revenue** and purchase more frequently.
2. The **26–35 age group** is the highest spending demographic.
3. **Married women** are the most valuable customer segment.
4. Top revenue-generating occupations:
 - IT Sector
 - Healthcare
 - Government Jobs
5. **Top performing states:** Uttar Pradesh, Maharashtra, Karnataka.
6. **Clothing, Electronics, and Food** dominate product category sales.
7. Customer demographics strongly influence purchase behavior.

Business Recommendations

1. Targeted Marketing

- Create gender-specific campaigns focused on women (especially married women).
- Build age-focused campaigns for the 26–35 segment.

2. Region-Specific Promotions

- Invest in digital marketing for top states: UP, Maharashtra, Karnataka.

3. Product Strategy

- Increase inventory for Clothing, Electronics, and Food categories.
- Promote top 10 bestselling products with bundled offers.

4. Customer Engagement

- Introduce loyalty benefits for married female customers.
- Develop festive “family bundle offers” for households.

5. Pricing & Discounts

- Offer limited-time festival discounts on high-performing categories.
- Provide exclusive membership pricing for working professionals.

Conclusion

The Diwali Sales Analysis reveals strong patterns in customer demographics, spending power, and product preferences. These insights help businesses make informed decisions about festive campaigns, marketing strategies, inventory planning, and targeting high-value customers.

With data-driven decision-making, businesses can significantly enhance revenue, improve customer satisfaction, and optimize their Diwali sales performance.